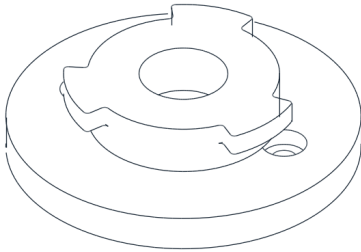


ARM-450 — Quick-Change End Effectors

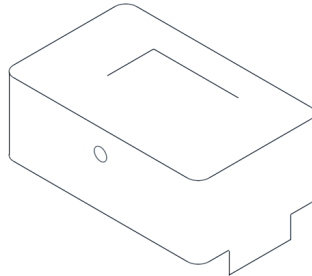
Two tools, one coupling · 15/15 interface checks pass · preflight stage 8

ARM-450 — quick-change end effectors · three-lug bayonet, one thumbscrew

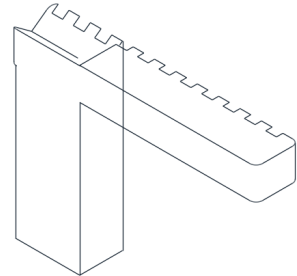
QUICK-CHANGE ADAPTER — stays on J6
48 × 48 × 18 mm



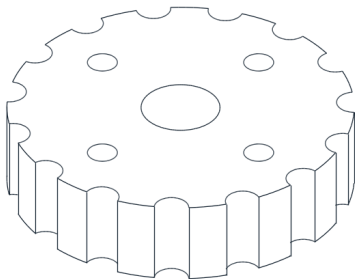
GRIPPER BODY
54 × 34 × 26 mm



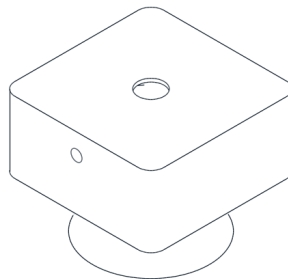
GRIPPER JAW ×2 (V-groove)
46 × 8 × 34 mm



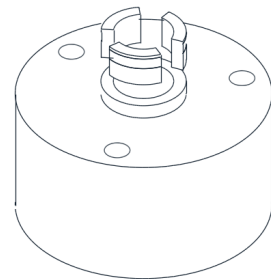
PINION Ø20
22 × 22 × 6 mm



DOCKING PROBE
46 × 46 × 38 mm



DOCK TARGET — bench test
34 × 34 × 28 mm



The coupling

A three-lug bayonet plus one M3 thumbscrew. Push on, twist 30°, nip the screw. Off in about five seconds with no tools.

Why not bolts. The J6 face has only three bolt holes — the servo pocket occupies the fourth quadrant, which the geometry audit established. Asking you to start three blind M3 screws under a wrist every time the tool changes is the wrong answer to a tool-change problem. The three bolts are used once, to fix the adapter, and never touched again.

part	size	mass	stays / changes
tool_adapter	48 × 48 × 18	14.4 g	bolts to J6 once
tool_gripper + 2 jaws + pinion	54 × 34 × 26	49 g + 9 g SG90	changes
tool_dock	46 × 46 × 31	48.0 g	changes
dock_target	34 × 34 × 31	12.6 g	bench test, not on the arm
dock_target_sealed	34 × 34 × 31	12.5 g	the same port with an O-ring groove, for fluid transfer

Verified against the real J6 face: bolt angles 160 / 250 / 340° match exactly, Ø30.00 bolt circle matches, pilot spigot 0.30 mm total clearance in the Ø10 bore, and both tool sockets take the male spigot at 0.30 mm per side.

Mass — a tool is payload, not arm

The arm is 1365 g against a 1450 g hard ceiling, with 85 g spare. Neither tool fits in that, and neither needs to. Every load analysis in this project assumed 300 g at the TCP, so a tool is payload.

Heaviest configuration — adapter + gripper + servo — is 75 g, a quarter of the allowance. Only one tool is fitted at a time.

Tool 1 — parallel-jaw gripper

One SG90 (9 g) drives a Ø20 pinion between two racks. 31.4 mm of opening per jaw over half a servo turn.

Parallel, not scissor. A scissor jaw grips a cylinder at two points that migrate as it closes — which is how small parts get squeezed out of one. Parallel faces stay square through the whole stroke.

The jaws carry a 90° V-groove. A flat jaw touches a cylinder on one line per side, so the part is free to roll out sideways under load. A V touches on two lines per jaw, four in total, and locates a round part on its own axis whatever its diameter.

Two defects the interface check caught here, that a render would not have. At the original Ø16 pinion the rack pitch line sat 1.20 mm outside the pinion PCD — the teeth never touched — and the half-turn stroke was 25.1 mm against a 30 mm design opening. Both are now correct: pitch line +0.00 mm on the PCD, stroke 31.4 mm. A gear mesh is exactly the thing that looks fine in a picture and does nothing in plastic.

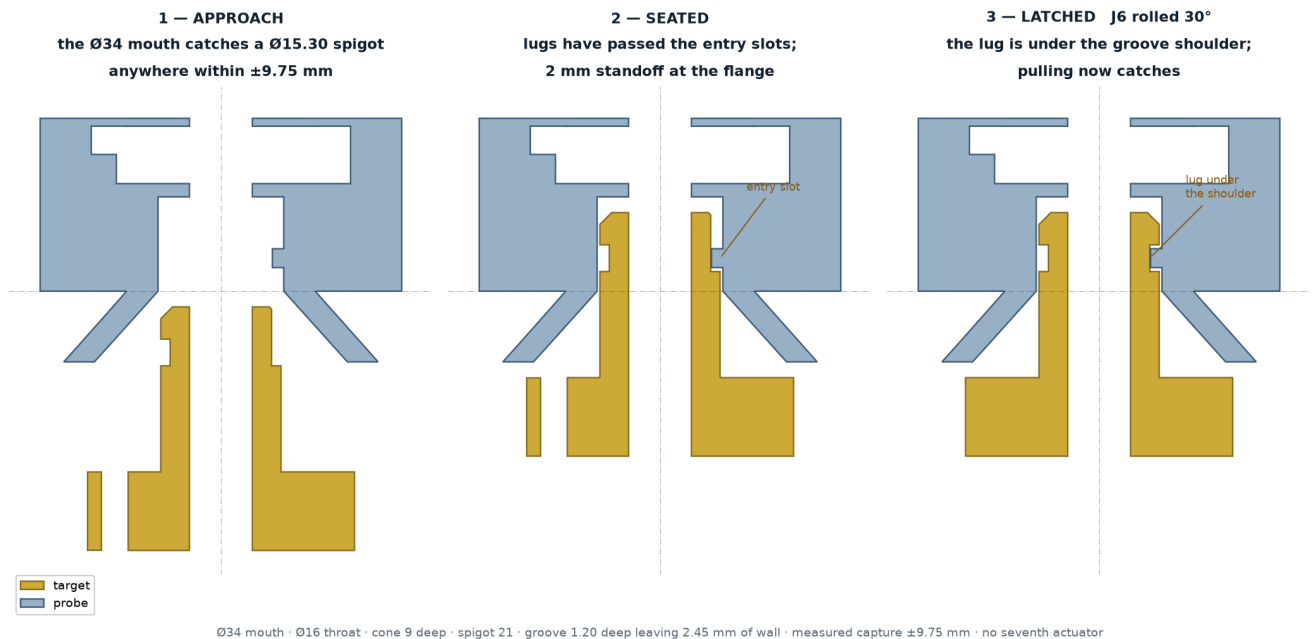
Tool 2 — MASCOT docking probe

No seventh actuator. The latch is a bayonet and J6 turns it. Insert the probe, roll J6 by 30°, the lugs are captured. Reverse to undock. J6 has ±175° of travel and does nothing during a docking approach, so the motion is free.

Capture tolerance is the number that matters. The cone is a Ø34 mouth over a Ø16 throat, and the measured envelope — probe and target as exported meshes, swept sideways until they foul — is ±9.75 mm at the mouth. Expected end-to-end arm precision is around 1.4 mm once servo backlash is included, so the cone carries about 7× the error the arm can actually make. That margin is deliberate: docking is the one task where a miss is expensive.

A Ø8 fluid pass-through runs the full length of both probe and target, for the refuelling line.

MASCOT DOCKING LATCH — section through a lug centre



The latch is the one feature here that no isometric can show — it is inside a bore. That is a large part of why it was wrong for as long as it was. Read left to right: the cone catches the spigot, the lugs pass down through the entry slots, and rolling J6 by 30° puts each lug under a shoulder of target material. Pull now and it catches.

What was wrong with the first docking design

Three defects, all found on 2026-08-22 by asking the geometry instead of looking at the render. Every one of them would have survived to the print bed.

1 — the latch did not exist. The three latch lugs were placed inside the capture cone at a height where the bore had already opened to Ø25.8, so they floated 5.5 mm clear of any wall. tool_dock came out of CadQuery as four solids: the probe, and three loose 19 mm³ crumbs. A .union() of shapes that do not touch does not fail — it returns a compound, and every downstream step (render, drawing, mass) treats it as one part.

2 — the target's spigot was cut in half. The capture groove was cut from r3.10 to r6.40 through a wall that ran r4.00 to r5.65 — a 3.30 mm groove in a 1.65 mm wall. That is not a groove, it is a parting cut: it severed the tip from the flange, and the three entry slots then chopped what remained into three loose arcs.

3 — the cone was deeper than the spigot was long. Even with both parts whole, they could not have mated. The capture cone was 16 mm deep and the spigot 10 mm tall, so the cone mouth grounded on the Ø34 flange while the tip was still 6 mm short of the throat. Probe and target collided at every insertion depth and every roll angle.

The fixes: throat Ø12 → Ø16 and mouth Ø30 → Ø34 (which keeps the ±9 mm capture while giving the spigot a 2.45 mm wall at the groove root); cone 16 → 9 mm deep; spigot 10 → 21 mm, so the post out-reaches the cone with 2 mm of standoff when latched; and the lugs now grow inward from the throat wall, overlapping it by 1 mm so the union has real material to fuse to.

The mate is now simulated, not asserted. check_tool_fit.py drives the exported probe

mesh onto the exported target mesh and checks all five states:

state	expected	result
6 mm short, slots aligned	clear	clear
seated, slots aligned	clear	clear
rolled 30° to latch	clear	clear
latched, pulled 2 mm	must catch	catches
rolled back, withdrawn	must release	releases

None of these three defects is visible in a render. Two of them look right in a render. The only reason they were found is that the parts were asked a question — how many solids are you, and do you actually mate — that a picture cannot answer.

dock_target is the passive port, printed so the interface can be tested on a bench before anything flies or floats. Bolt it down, fit the probe, drive in by hand and roll J6 30°. It should latch and release repeatably. That test needs no fuel, no MASCOT hardware and no flight software.

Printing

Not on the critical path for the arm — print these after it.

order	part	qty	why this order
1	tool_adapter	1	small; fits to J6 and stays
2	tool_dock + dock_target	1 + 1	lets the docking interface be bench-tested without the arm
3	tool_gripper, gripper_jaw ×2, gripper_pinion	1/2/1	needs the SG90 to be useful

Same PLA+CF spool as the arm. Print the pinion and jaw racks at 100 % infill — the teeth are small and sparse infill leaves them hollow.

To buy: one SG90 micro servo, two M3 × 16 thumbscrews.

Verifying

```
python3 cad/end_effector.py    # rebuild all six parts
python3 check_tool_fit.py      # 15 interface checks
python3 preflight.py           # full gate, tools are stage 8
```